

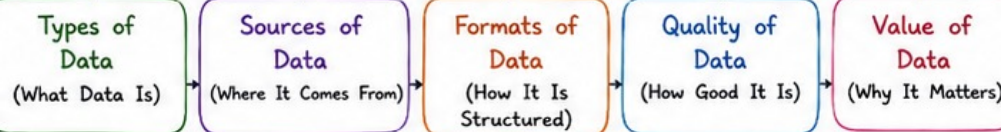
CONCEPT MAP

FACETS OF DATA

Understanding the Different Dimensions, Types and Roles of Data

Exploring the nature, variety and significance of data in decision making, analytics and real-world applications

THESE FACETS WORK TOGETHER



1. TYPES OF DATA

Data can be categorized based on its nature and characteristics.

TYPES

- Qualitative Data – Descriptive, non-numeric (e.g., names, reviews)
- Quantitative Data – Numeric, measurable (e.g., age, sales)
- Discrete Data – Countable, whole numbers
- Continuous Data – Measurable on a continuum (e.g., height, time)
- Categorical Data – Grouped into categories
- Ordinal Data – Ordered categories (e.g., rankings)

2. SOURCES OF DATA

Data is collected from various origins and channels.

SOURCES

- Internal Sources – Within an organization (e.g., databases, logs, transactions)
- External Sources – Outside the organization (e.g., surveys, APIs, public data, social media)
- Primary Data – Collected first-hand for a specific purpose
- Secondary Data – Already collected by someone else



3. FORMATS OF DATA

Data can be stored and represented in different formats.

FORMATS

- Structured Data – Organized in rows and columns (e.g., tables)
- Semi-structured Data – Has tags or markers (e.g., JSON, XML)
- Unstructured Data – No fixed format (e.g., text, images, videos)



4. QUALITY OF DATA

The usefulness of data depends on its quality.

DIMENSIONS

- Accuracy – Correct and error-free
- Completeness – No important data is missing
- Consistency – Uniform across systems and time
- Timeliness – Available when needed
- Validity – Conforms to defined rules and formats



5. VALUE OF DATA

High-quality data delivers meaningful value.



Informs Better Decisions



Drives Innovation



Improves Efficiency



Enhances Customer Experience



Identifies Risks and Opportunities

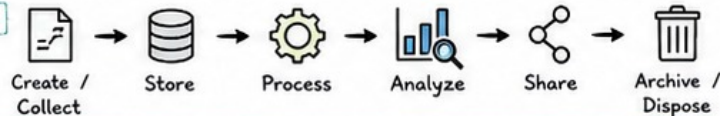


Supports Strategic Planning

6. DATA LIFECYCLE (A KEY PERSPECTIVE)

Data goes through a lifecycle from creation to disposal.

STAGES



7. ROLE OF DATA

Data plays a central role across industries and functions.

ROLES



Supports Analytics & AI/ML



Enables Automation



Supports Reporting & Visualization



Drives Personalization



Ensures Compliance & Governance

8. CHALLENGES IN MANAGING DATA

Organizations must address common data challenges.

CHALLENGES



Data Overload



Data Privacy & Security



Data Silos



Data Quality Issues



Integration Complexity



Cost of Storage & Processing



Skill Gaps

9. BEST PRACTICES

Follow these practices to leverage data effectively.

PRACTICES



Define Clear Objectives



Ensure Data Governance



Maintain Data Quality



Use the Right Tools



Foster a Data-Driven Culture

★ **SUMMARY :** The facets of data—its types, sources, formats, quality and value—work together across the data lifecycle to drive insights, support decisions and create impact. Managing these facets effectively leads to trustworthy, useful and responsible data.

